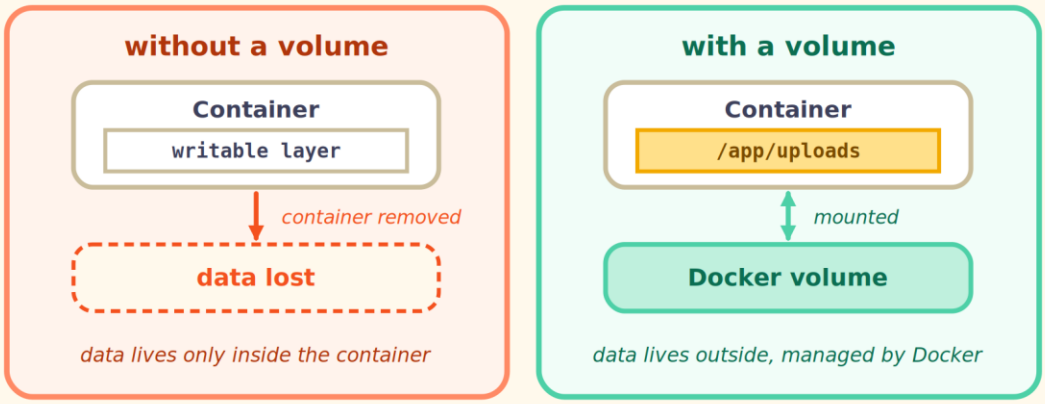
Docker Storage & Compose

1. Why do we need storage?

By default, data saved inside a container may be lost when the container is removed. Docker provides persistent storage so that data can live outside the container's writable layer.



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2. Types of Docker storage



- **Volumes:** managed by Docker and used to persist data independently of any container. Recommended for databases and application data.
- **Bind mounts:** map a specific folder or file from the host machine into the container.

- **Container writable layer:** data stored in the container's own filesystem. It's not reliable for long-term persistence.

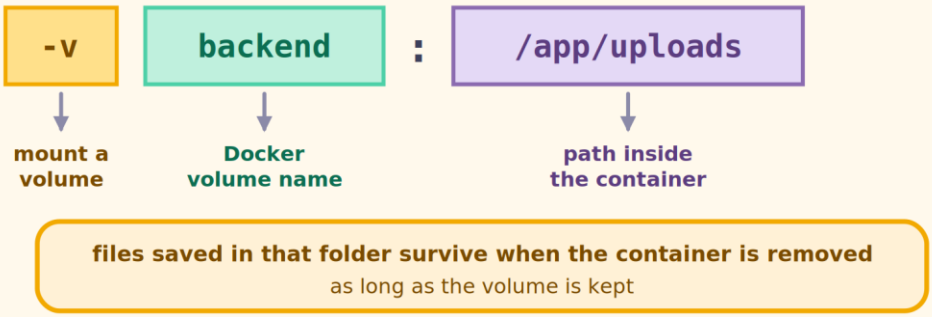
---

3. Essential storage commands

```
docker volume ls ~ list volumes
docker volume create backend ~ create a volume
docker volume inspect backend ~ inspect a volume
docker run -d --name server -p 8080:8080 -v backend:/app/uploads server
```

Understanding -v

Files stored in the container path persist when the container is removed, as long as the volume is kept.

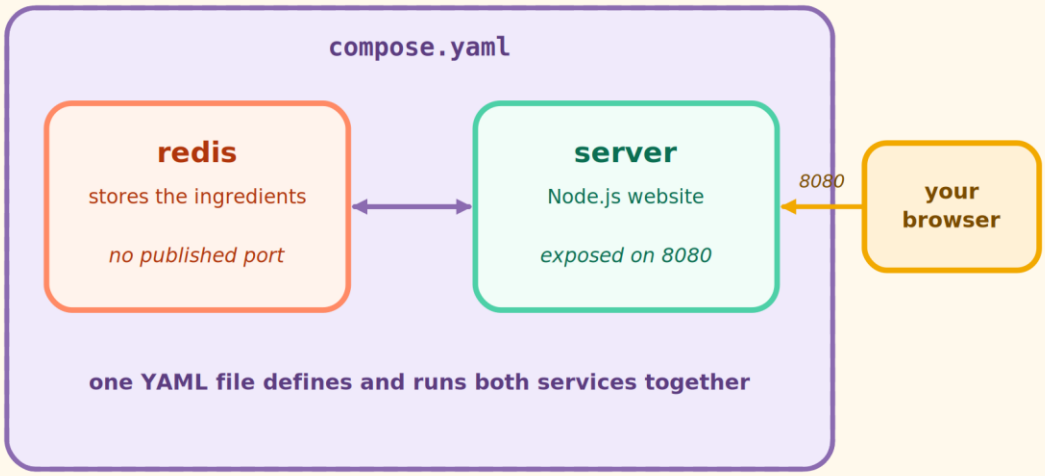


~ removing a container does not normally remove its named volume. To delete a volume, run docker volume rm backend after no container is using it.

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4. What is Docker Compose?

Docker Compose defines and runs multiple related containers using a single YAML file, usually named compose.yaml. In this project it runs two services: server, the Node.js website exposed to the host on port 8080, and redis, which stores the ingredients and is not directly exposed to the host.



Essential Compose commands

Run these from the directory that contains compose.yaml:

```
docker compose up --build -d   ~ build images and start services
docker compose ps              ~ list services and their status
docker compose logs -f server  ~ follow the server logs
docker compose logs -f redis   ~ follow the Redis logs
docker compose down            ~ remove containers and networks
docker compose down -v         ~ also remove named volumes
```

Public

Careful with -v

docker compose down -v deletes the Compose-managed named volumes and the data stored in them. Don't use it if you want to keep the uploaded images and the Redis data.

